

mdstats 0.20.216a0 - TRAIN2 FP32 repeatability diagnostic hotfix

Scope

This release adds diagnostic measurement only. It does not change any e3nn/CuEq parity tolerance or scientific convergence criterion.

Diagnostic

- Run 10 repeated e3nn and 10 repeated pure-CuEq TRAIN2 evaluations for FP32 phase-separated CuEq doctor checks.
- Report paired **E_{max}**, **F_{max}**, **F_{rmse}**, force p99/p99.9, force-component count above **1e-5**, **S_{max}**, **D_{max}**, and selection identity for every repetition.
- Report aggregate min/median/p90/max for e3nn-self **F_{max}**, CuEq-self **F_{max}**, cross-backend **F_{max}**, **F_{rmse}**, p99, and p99.9.
- Print PyTorch/cuDNN/CUBLAS determinism state.
- Persist the content-addressed non-authorizing diagnostic record in campaign state.

Unchanged authority

TRAIN2 FP32 remains **rtol=1e-5**, **atol=1e-5**. Generic source/DATA6 FP32 remains **rtol=1e-5**, **atol=1e-6**; FP64 remains **rtol=1e-10**, **atol=1e-12**. FINAL-GPU1 HF2 is held archival until the workstation repeatability values are reviewed.